

Week 3, Day 1 — student handout

Umesh Indrajith · Keep this sheet. Two labs today. SMS still starts week 4.

TODAY'S OUTCOMES

- if, short statement, switch (including tagless).
- for / range / break / continue.
- Slices: len, cap, make, append, backing-array gotcha.
- Maps: make, comma-ok, range (order not promised).
- Functions enough to ship helpers; return x, err as a copyable pattern.
- Green tests for Contains, Unique, Freq.

FOLDER FOR TODAY

```
mkdir -p ~/epic-go/week03/slicekit  
cd ~/epic-go/week03
```

VS Code: File → Open Folder → ~/epic-go/week03. gofmt on save is still required.

CONTROL FLOW

```
if n := len(name); n == 0 {  
    fmt.Println("empty")  
}  
  
switch day {  
case "Sat", "Sun":  
    fmt.Println("weekend")  
default:  
    fmt.Println("weekday")  
}  
  
for i, v := range courses {  
    fmt.Println(i, v)  
}
```

SLICES

```
s := make([]string, 0, 8)  
s = append(s, "SE")           // always assign the result  
  
a := []int{10, 20, 30, 40}  
b := a[:2]  
b = append(b, 99)           // may change a[2] — shared backing array
```

Slice header = pointer + length + capacity. Need independence? Copy into a new backing array.

MAPS

```
count := make(map[string]int)
count["SE"]++
v, ok := count["SE"]      // comma-ok: missing vs present
if !ok { /* not found */ }
delete(count, "SE")
```

Writing to a nil map panics. Range order is randomised — do not assert print order in tests.

LAB PACKAGE

```
cd ~/epic-go/week03/slicekit
go mod init epiclearn/week03
# implement Contains, Unique, Freq + tests
go test
```

WORDS TO REMEMBER

Term	Meaning today
Backing array	Storage underneath one or more slices
cap	How far append can grow without reallocating
Comma-ok	<code>v, ok := m[k]</code> — was the key present?
Tagless switch	<code>switch { case cond: }</code>
–	Blank identifier — intentionally unused

IF IT BREAKS

- `if score` → need a bool comparison
- `append ignored` → assign: `s = append(s, x)`
- `nil map write` → make first
- `flaky map test` → assert keys, not print order
- `go test no module` → `go mod init`

HOMEWORK (BEFORE NEXT CLASS)

1. Three short answers in `notes.md` (append, backing array, comma-ok).
2. Coding in `homework/`: Contains, Unique, Freq, Filter, Max, Reverse, Intersection, Grade — green `go test`.
3. Starter stubs: `docs/week-03-day-01/starter/homework/`. Full sheet: `homework.md`.